

PROJECT REPORT

Customer Analytics & Relational SQL Pipeline

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1. Executive Summary

This report outlines the development and analysis of the RetailPulse e-commerce dataset. The project aimed to transform raw customer data into actionable business intelligence to optimize revenue, improve retention, and refine marketing strategies. The analysis processed 5,000+ records and uncovered critical insights into customer demographics, revenue drivers, and product sentiment.

2. Business Problem & Objectives

The retail organization faced challenges in understanding customer behaviour and marketing efficiency. The analysis addressed six key business problems:

1. Low Customer Retention : Distinguishing between frequent and one-time buyers .
2. Ineffective Marketing : Evaluating the real impact of discounts and promo codes .
3. Product Demand : Identifying high-performing categories and revenue generators .
4. Payment Behaviour : Analyzing preferred transaction methods .
4. Seasonal Variation : Mapping sales trends across seasons .
5. Customer Segmentation : Profiling customers as new, returning, or loyal .

3. Technical Methodology

The project followed a full-stack data analysis lifecycle :

1. Data Loading & Wrangling (Python/Pandas) : Handled missing values using domain-specific logic (e.g., product-level mean for ratings, mode for clothing sizes) . Standardized columns for SQL compatibility . Removed duplicates via `Customer\_ID` .
2. Database Management (SQL Server) : Exported cleaned data to SQL Server . Executed advanced queries (Window Functions, CTEs) to identify high-value segments and revenue trends .
3. Reporting & Visualization : Synthesized results into executive-ready reports for business decision-making .

4. Key Analytical Insights

1. Revenue Drivers: The Electronics category is the primary revenue engine (\$480,210.81).
2. Demographic Power : Customers aged 51+ are the highest contributors (\$399,214.36).
3. Discount Impact : Analysis proved that while discounts increase average purchase value (from \$180.19 to \$217.69), they must be applied strategically .
4. Satisfaction Gap : High-revenue Electronics items (Phones, Laptops) have lower satisfaction scores, suggesting a need for quality improvements .
5. Customer Loyalty : Identified 3,116 "Loyal Customers," representing a core group for future subscription programs .

5. Analysis of Key Business Inquiries

This section provides SQL-based evidence for core business questions.

1. Q: Which category generates the highest revenue?

Finding**: Electronics generates the highest revenue at \$486.37K .

	category	total_revenue
▶	Electronics	480210.81
	Accessories	194456.78
	Clothing	189891.4
	Footwear	111653.33
	Outerwear	18558

2. Q: Are discounts actually increasing purchase value?

Finding**: Yes, discounts increase average purchase value from \$182.24 to \$219.

	discount_applied	total_revenue	avg_purchase
▶	No	502562.72	180.19
	Yes	492207.59	217.69

3. Q: What is the total revenue generated by male and female customers?

Finding**: Male customers generated \$390.96K (39.33%) and Female customers generated \$352.06K (35.41%)

	gender	total_revenue
▶	Male	392443.71
	Female	352269.39
	Other	250057.21

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4. Q: Which customer used a discount but still spent more than the average purchase amount?

Finding**: Multiple customers (e.g., Customer IDs 4256, 4289) utilized discounts while maintaining a purchase amount significantly above the average, indicating high-value discount sensitive segments.

customer_id	purchase_amount	discount_applied
4977	1499.76	Yes
4055	1493.56	Yes
4226	1472.22	Yes
4600	1460.51	Yes
4662	1459.59	Yes
4408	1456.31	Yes
5091	1451.07	Yes
4573	1441.51	Yes
4940	1439.75	Yes
4079	1438.53	Yes

5. Q: Which are the top/bottom 5 products with the highest average review rating?

Finding**: Top 5: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.79). Bottom 5: Phone (2.94), Headphones (3.08), Watch (3.10), Bag (3.17), Laptop (3.19) .

item_purchased	avg_rating
Gloves	3.86
Sandals	3.83
Boots	3.81
Hat	3.8
T-shirt	3.79

item_purchased	avg_rating
Phone	2.92
Watch	3.08
Headphones	3.11
Laptop	3.15
Bag	3.16

6. Q: Average purchase: Standard vs Express shipping?

Finding**: Express shipping yields \$358.12 total revenue across 1199 orders, while Standard shipping yields \$340.57 across 1201 orders.

shipping_type	total_orders	avg_order_value	total_revenue
Express	1195	354.72	426730.45
Standard	1212	335.58	410744.86
Free Shipping	674	60.46	41231
Store Pickup	650	59.89	39226
2-Day Shipping	626	60.91	38675
Next Day Air	643	58.62	38163

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7. Q: Do subscribed customers spend more?

Finding : Yes, subscribers have an average spend of \$270.58 compared to \$165.72 nonsubscribers.

	subscription_status	total_users	avg_revenue	total_revenue
▶	No	3473	163.89	569195.3
	Yes	1577	269.86	425575.02

8.Q: Compare average spend and total revenue between subscribers and non-subscribers?

Finding : Subscribers (1,579 customers) generated \$427.24K total revenue, while Nonsubscribers (3,421 customers) generated \$566.94K total revenue; however, subscribers maintain a significantly higher average spend per transaction.

	subscription_status	total_users	avg_revenue	total_revenue
▶	No	3473	163.89	569195.3
	Yes	1577	269.86	425575.02

9. Q: Top 5 products with highest discount usage %?

Finding**: Laptop (51.75%), Phone (51.46%), Headphones (51.20%), Hat (50.00%), and Watch (49.68%).

	item_purchased	total_sold	times_discounted	discount_usage_percent
▶	Laptop	140	74	52.86
	Headphones	166	86	51.81
	Sneakers	142	72	50.70
	Phone	170	86	50.59
	Hat	153	77	50.33

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10. Q: Segment customers into new, returning, and loyal based on their total number of previous purchases?

Finding**: Loyal Customers (3085), Returning Customers (1354), New Customers (561)

	customer_segment	customer_count
▶	Loyal Customers	3116
	Returning Customer	1373
	New Customer	561

11. Q: What are the top 3 most purchased products within each category?

Finding**: Accessories: Jewelry (171), Bag (164), Belt/Sunglasses (161). Clothing: Shirt (315), Blouse (171), Pants (171). Electronics: Phone (171), Headphones (166), Watch (157). Footwear: Shoes (303), Sandals (160), Sneakers (145). Outerwear: Jacket (163), Coat (161)

	category	item_purchased	purchase_count	rnk
▶	Accessories	Jewelry	180	1
	Accessories	Sunglasses	163	2
	Accessories	Bag	162	3
	Clothing	Shirt	370	1
	Clothing	Blouse	169	2
	Clothing	Pants	168	3
	Electronics	Phone	170	1

12. Q: Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Finding**: Yes, 1190 Repeat Buyers have active subscriptions, whereas only 389 Normal Buyers are subscribed.

	buyer_type	subscription_status	customer_count	percentage
▶	Normal Buyers	No	662	62.75
	Normal Buyers	Yes	393	37.25
	Repeat Buyers	No	2811	70.36
	Repeat Buyers	Yes	1184	29.64

13. Q: What is the revenue contribution of each age group?

Finding**: 51+ (\$399.3K, 40.2%), 36-50 (\$266.7K, 26.8%), 26-35 (\$177.4K, 17.8%), 18-25 (\$150.7K, 15.2%)

	age_group	total_revenue
▶	18-25	151102.15
	51+	399214.36
	36-50	266355.41
	26-35	178098.38

6. Strategic Recommendations

1. **Targeted Marketing** : Focus promotional activity for the 51+ age demographic .
2. **Retention Programs** : Launch loyalty incentives for the 3,116 identified loyal customers.
3. **Product Quality** : Address the discrepancy between high sales and low ratings for Electronics .
4. **Operational Efficiency** : Optimize inventory based on the high performance of items like Jewelry, Shirts, and Shoes .

7. Conclusion

The RetailPulse analysis successfully bridged the gap between technical data processing and business strategy . The project provided actionable insights that directly answer key organizational questions and offer a clear path for revenue optimization .

